

GreenVision.AI

AI-Powered Environmental Intelligence for Cities, Industries & People

Turning environmental data into green action

Team ID: I4I 633 | Team: Hacksmith | Leader: Nayana N | PS ID: PS-I04 (Software)

J.S.S. Academy of Technical Education, Bengaluru

Smart India Hackathon 2026 | Problem Statement Category: Software

PROBLEM & SOLUTION

THE PROBLEM

- Indian cities lose 1.5-3% tree cover per decade (ISFR 2023)
- No real-time tool to measure urban green cover from imagery
- Municipalities plan plantations without data-driven priority mapping
- Industrial sites lack green buffer compliance assessment
- Citizens cannot assess or act on local environmental data

OUR SOLUTION

- AI-GIS platform: upload imagery -> environmental diagnosis
- 3 distinct workspaces: Municipal / Industrial / Citizen
- ML models: canopy segmentation, tree detection, scene classification
- Location-conditioned species engine (Open-Meteo weather/AQI/soil)
- Conversational AI assistant grounded in real data

7.92%

Canopy Cover

156

Trees Detected

1,026

Trees Needed

2.2 t/yr

CO2 Offset

35/35

Tests Passed

THREE WORKSPACES

One platform - three tailored experiences for every stakeholder

MUNICIPAL

- Upload city/ward imagery
- AI canopy detection & analysis
- 60% green cover target tracking
- Planting gap quantification
- GIS map with priority zones
- Carbon sequestration & oxygen output
- Exportable assessment report

INDUSTRIAL

- Site green buffer assessment
- Regulatory compliance status
- Buffer zone planning support
- Pollution-tolerant species advisory
- Green investment estimates
- ESG reporting metrics
- Maintenance cost projection

CITIZEN

- Location-aware environmental profile
- Interactive Leaflet map
- Live weather / AQI / soil data
- AI chatbot advisor
- Natural conversation interface
- Green Points rewards system
- Community leaderboard

GREENVISION AI - CONVERSATIONAL ASSISTANT

Not a keyword matcher. A context-aware environmental advisor.

USER

"Hi, my name is Nayana"

GREENVISION AI

"Hey Nayana! I'm GreenVision, your environmental assistant. Ask me about green cover, trees, or your area's air quality."

USER

"What trees are good for my area?"

GREENVISION AI

"Based on your location in Bengaluru and its climate/soil profile, top match: 1. Neem (82/100) - drought-tolerant, air-purifying..."

USER

"Give me another option"

GREENVISION AI

"Here are other options suited to Bengaluru: 1. Pongamia (75/100) - hardy canopy native; 2. Indian Laurel (73/100)..."

USER

"How many trees do I need?"

GREENVISION AI

"Your ward currently has 156 trees (7.92% canopy). About 1,026 additional trees to reach the 60% target."

Session memory | Name-aware greetings | Follow-up context | Species exclusion | Cost follow-ups | Honest fallback

ORGANIZATION ECOSYSTEM

Every contribution is attributed - individuals build reputations, organizations climb rankings

CONTRIBUTION SYSTEM

- 6 org types: Individual, Company, College, School, NGO, Community
- Organization stored with each contribution record
- Backend aggregates points by organization
- 4 action types earn Green Points: plant, adopt, verify, review

LEADERBOARD

- Individual rankings with podium animation for top 3
- Organization aggregation: e.g. Infosys (24 participants, 8,700 pts)
- Dedicated tabs: Companies, Colleges, NGOs, Communities
- Transparent scoring visible to all participants

"Green Points are a participation score, not a CO2/O2 measurement."

TECHNICAL ARCHITECTURE

End-to-end pipeline from imagery to actionable insight



FRONTEND	React 19 Vite TailwindCSS React-Leaflet
BACKEND	Flask Python 3.11 PyTorch 2.2
ML MODELS	YOLOv8 (tree detection) U-Net (canopy segmentation) Scene classifier
DATA	Open-Meteo (weather, AQI, soil, elevation) Nominatim (geocoding)
STORAGE	Flat JSON (demo-grade) localStorage for identity

IMPACT & FEASIBILITY

IMPACT

- Helps municipalities understand environmental health of every neighborhood
- Identifies areas with low green cover, high heat, and poor air quality
- Enables data-driven plantation decisions - right place, right species
- Creates community engagement through Green Champions and leaderboards

FEASIBILITY

- Uses readily available satellite/drone imagery and open datasets
- Built on scalable, proven AI, GIS, and web technologies
- No specialized hardware dependency - runs on commodity infrastructure
- Supports phased deployment from ward level to full city scale

FUTURE ROADMAP & REFERENCES

FUTURE ROADMAP

- Pollutant dispersion model for industrial mode
- Historical trend analysis (multi-temporal comparison)
- Account system with verified identity
- Mobile app for field verification
- Integration with municipal GIS databases

REFERENCES

- Smart Cities Mission (MoHUA)
- ISRO Bhuvan Geoportal
- Forest Survey of India (ISFR)
- Open-Meteo API
- YOLOv8 / Ultralytics

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